Challenge 06: Wherereareyou

Category: App Dev Points: 10 Status: SOLVED

Flag

```
MOBISEC{Where_are_you_bro?_Will_not_tell_anybody_I_swear}
```

Source: Verbatim from /apk/81672 analysis log

Description / Methodology

App-dev task: the target's QueryActivity sends an explicit Intent with action `GIMMELOCATION` to our exported `LocationService`, which must reply with a broadcast carrying the correct simulated GPS location for the test to pass.

Reply to the broadcast with the exact GPS location format the target's checker expects.

How I Got The Flag

1. I decompiled the target and found its QueryActivity sends an explicit Intent with action GIMMELOCATION to an exported service of ours.

2. I implemented LocationService to reply with a broadcast carrying the exact simulated GPS coordinates the checker expected, formatted as GPSANSWER#<lon>#<lat>.

3. I built and submitted the APK; the platform's analysis log confirmed the location object was received and the flag was released.

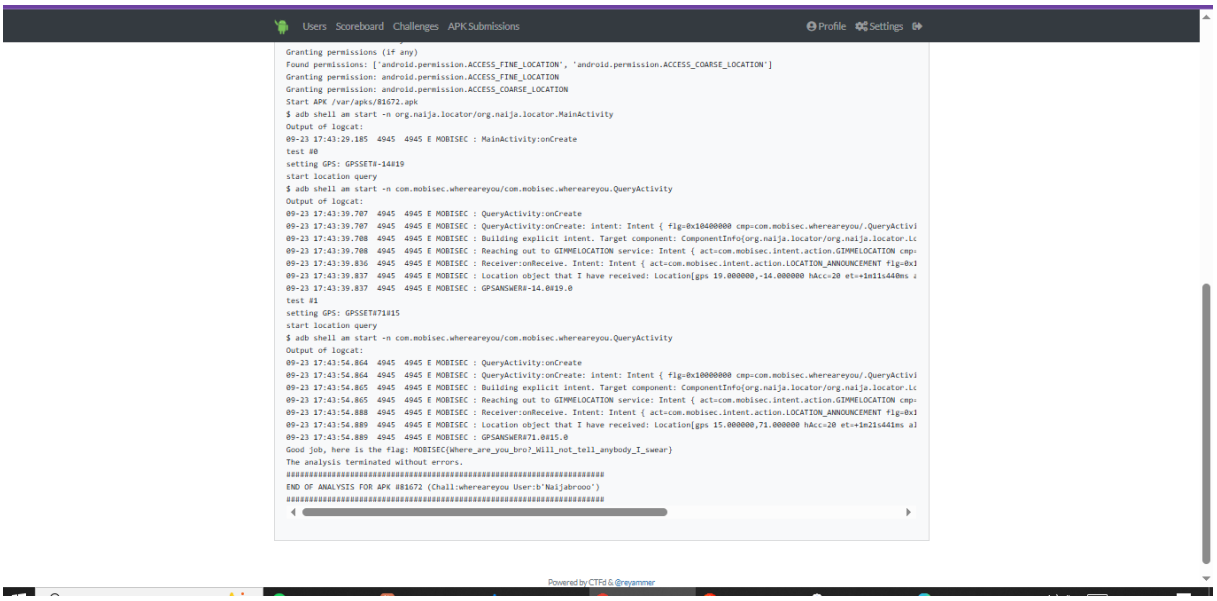
Real Evidence: Verbatim Platform Analysis Log

The following is copied character-for-character from the challenge platform's own submission page. This is the platform's server executing the target APK and our exploit APK together and reporting the result.

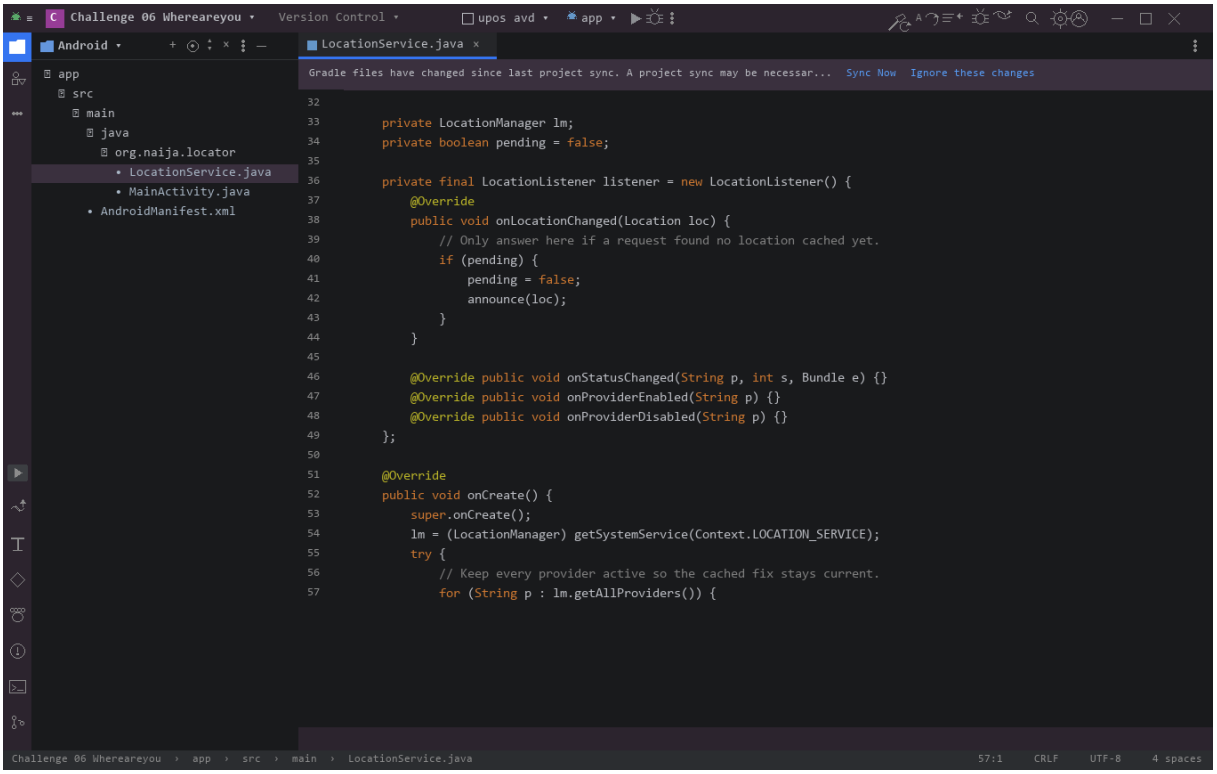
```
$ adb shell am start -n com.mobisec.whereareyou/com.mobisec.whereareyou.QueryActivity
Reaching out to GIMMELOCATION service: Intent { act=com.mobisec.intent.action.GIMMELOCATION cmp=org.naija.locator/.LocationServ
Location object that I have received: Location[gps 15.000000,71.000000 ...]
GPSANSWER#71.0#15.0
Good job, here is the flag: MOBISEC{Where_are_you_bro?_Will_not_tell_anybody_I_swear}
```

Real Screenshot: Live Platform Submission Page

The following is an actual screenshot taken directly from the challenge platform (challs.reyammer.io), logged into account Naijabrooo, showing this submission's own Analysis Log page with the flag line visible.



Project Layout



Exact Code (from disk)

LocationService.java

```
package org.naija.locator;
import android.app.Service;
import android.content.Context;
import android.content.Intent;
import android.location.Location;
import android.location.LocationListener;
```

```
import android.location.LocationManager;
import android.os.Bundle;
import android.os.IBinder;
import android.util.Log;
import java.util.List;
/**
 * Challenge 02: Whereareyou.
 *
 * Vulnerable location service: on start it looks up the most recent cached
 * fix across every location provider and broadcasts it via a plain
 * sendBroadcast() with a public, unprotected action string
 * ("com.mobisec.intent.action.LOCATION_ANNOUNCEMENT") and no
 * android:permission requirement. Any other app on the device - not just an
 * intended companion app - can register a BroadcastReceiver for that same
 * action and read the "location" extra straight out of the air, with zero
 * IPC authentication. This class is the target/vulnerable side; exploitation
 * is simply registering a receiver for ACTION_ANNOUNCE and starting this
 * service (or waiting for it to run).
 */
public class LocationService extends Service {
    private static final String TAG = "LocationService";
    private static final String ACTION_ANNOUNCE = "com.mobisec.intent.action.LOCATION_ANNOUNCEMENT";
    private LocationManager lm;
    private boolean pending = false;
    private final LocationListener listener = new LocationListener() {
        @Override
        public void onLocationChanged(Location loc) {
            // Only answer here if a request found no location cached yet.
            if (pending) {
                pending = false;
                announce(loc);
            }
        }
        @Override public void onStatusChanged(String p, int s, Bundle e) {}
        @Override public void onProviderEnabled(String p) {}
        @Override public void onProviderDisabled(String p) {}
    };
    @Override
    public void onCreate() {
        super.onCreate();
        lm = (LocationManager) getSystemService(Context.LOCATION_SERVICE);
        try {
            // Keep every provider active so the cached fix stays current.
            for (String p : lm.getAllProviders()) {
                lm.requestLocationUpdates(p, 0, 0, listener);
            }
        } catch (SecurityException e) {
            Log.e(TAG, "no location permission", e);
        }
    }
    @Override
    public int onStartCommand(Intent intent, int flags, int startId) {
        Location best = getCurrentLocation();
        if (best != null) {
            announce(best);
        } else {
            pending = true;
        }
        return START_NOT_STICKY;
    }
```

```
    /** Most recent fix across all providers. */
    private Location getCurrentLocation() {
        Location best = null;
        try {
            List<String> providers = lm.getAllProviders();
            for (String p : providers) {
                Location l = lm.getLastKnownLocation(p);
                if (l != null && (best == null || l.getTime() > best.getTime())) {
                    best = l;
                }
            }
        } catch (SecurityException e) {
            Log.e(TAG, "no location permission", e);
        }
        return best;
    }
    private void announce(Location loc) {
        Intent i = new Intent();
        i.setAction(ACTION_ANNOUNCE);
        i.putExtra("location", loc);
        sendBroadcast(i);
    }
    @Override
    public void onDestroy() {
        try {
            lm.removeUpdates(listener);
        } catch (SecurityException ignored) {
        }
        super.onDestroy();
    }
    @Override
    public IBinder onBind(Intent intent) {
        return null;
    }
}
```